GGG–S–64
MARCH 26, 1964
SUPERSEDING
GGG–S–61a (in part)
November 3, 1942

FEDERAL SPECIFICATION

SAW, CROSSCUT, HAND, ONE-MAN AND TWO-MAN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers one-man and two-man crosscut saws.

1.1.1 *Federal specification coverage.*—This specification does not cover all types, classes, and tooth styles of crosscut saws, but only those most generally used by the Federal Government.

1.2 Classification.—Crosscut saws shall be of the following types, classes, and tooth styles as specified. (See 6.2.)

Type I.—One-man
  Style A teeth
  Style B teeth
Type II.—Two-man
  Class 1.—Wide straight back
    Style A teeth
    Style B teeth
  Class 2.—Medium wide straight back
    Style A teeth
    Style B teeth
  Class 3.—Narrow skewback
    Style A teeth
    Style B teeth
  Class 4.—Uniform width
    Style A teeth
    Style B teeth
  Class 5.—Bucking
    Style C teeth
  Class 6.—Felling
    Style C teeth

2. APPLICABLE SPECIFICATIONS, STANDARD AND OTHER PUBLICATIONS

2.1 There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

2.2 Standards.—The following Military Standard, of the issue in effect on date of invitation for bids, forms a part of this specification:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

2.3 Other publications. — The following publications, of the issues in effect on date of invitation for bids, unless otherwise stated, form a part of this specification:

American Trucking Association:
Motor Freight Classification Rules.
Consolidated Classification Committee:
Consolidated Freight Classification Rules.
Post Office Department:
Postal Regulations.

(Non-Government activities may obtain copies of Federal Specifications and Standards upon application accompanied by check, money order, cash, or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets SW, Washington

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25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Copies of the Index of Federal Specifications and Standards may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C. Prices may be obtained from the Index of Federal Specifications and Standards or from the GSA Regional Offices.

(Single copies of this specification and other product specifications required by non-Government activities for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards through departmental channels from the established sources of supply.

(Copies of Military standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

(Application for copies of Motor Freight Classification Rules should be addressed to the American Trucking Association, 1424 16th St., N. W., Washington 6, D. C.

(Application for copies of Consolidated Freight Classification Rules should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.

(Application for copies of Postal Regulations should be addressed to the Post Office Department, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Blades shall be of hot-rolled saw steel which can be heat treated and tempered to meet the hardness requirements of 3.3 and the bending test of 4.3.2. The steel shall be free from cracks, segregations, inclusions, seams, hammer marks, high spots, or other imperfections that may affect their serviceability.

3.1.2 Handles shall be of suitable hardwood thoroughly seasoned, straight grained, and free from any imperfections which may affect their serviceability or durability. Handles shall be finished smooth, stained and either wax coated or lacquered with a durable weather-resistant lacquer. The finish shall be transparent to the extent of not concealing the grain of the wood.

3.1.3 Unless otherwise specified, screws, medallions, and nuts shall be made of metal showing a Rockwell B hardness of not less than 60 and have a suitable rust-resisting coating.

3.2 Marking.—The blades shall be permanently marked with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined. The marking shall be applied in a manner that will preclude damage to the blade.

3.3 Hardness.—Blades which are spring set shall have a Rockwell C hardness of not less than 48 nor more than 52. Blades which are not set shall have a Rockwell C hardness of not less than 52 nor more than 58.

3.4 Teeth.—The teeth shall be similar to the conventional styles shown on figure 1.

3.4.1 Style A teeth shall have two cutters to one raker tooth. The cutter teeth shall be filed to a cutting point and beveled to an angle of approximately 45 degrees with the face of the saw and set to project laterally 0.005 to 0.008 inch from the plane of the face of the saw. Raker prongs shall be filed to a cutting edge square with the center plane of the thickness of the saw. The raker teeth shall not be set or swaged. The teeth shall be similar to style A teeth shown on figure 1.

3.4.2 Style B teeth shall have four lance cutting teeth for each raker tool; the teeth shall be filed and set in the same manner as style A teeth. The raker teeth shall not be set or swaged. The teeth shall be similar to style B teeth shown on figure 1.

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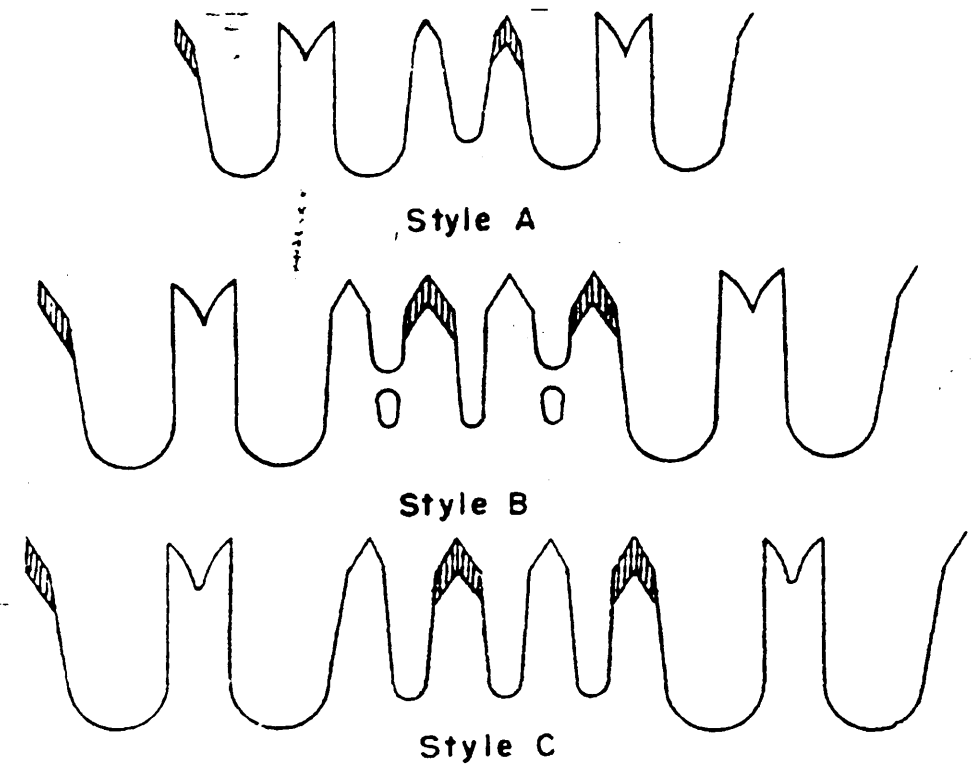


FIGURE 1.—Tooth styles A, B, and C.

3.4.3 Style C teeth shall have four nonperforated cutting teeth to each raker jointed to the same length along a smooth curve. The cutting teeth shall be filed to a cutting point and beveled at an angle of approximately 35 degrees with the face of the saws. Raker prongs shall be filed to a cutting edge square with the thickness of the saw. Cutting teeth shall not be set and rakers shall not be set or swaged. The teeth shall be similar to style C teeth shown on figure 1.

3.5 The tooth edge of the blade when measured just back of any set of teeth shall be uniform in thickness within 0.005 inch and shall be within a tolerance of plus or minus 0.004 inch of thickness of blade at tooth edge specified in individual tables of types and classes.

3.6 All blades shall be straight and true taper ground from the cutting edge to the taper ground on the back edge of the saw from the handle end to the center of the saw to provide clearance conforming to the dimensions shown in tables II to VI, inclusive.

3.7 The blades shall be hardened and heat-treated to permit filing, and type II saws shall withstand the bending test specified in 4.3.2.

3.8 Type I, one-man (see fig. 2). — The saw shall have a convex breast. Either straight or skewed-back blades are acceptable. The handle shall be of the one-horn or two-horn pattern with a large hand hole (suitable for use with heavy gloves) and shall be secured to the blade with not less than three screws and nuts. Saws shall have style A teeth or style B teeth, and the tapered clearance on the back of the blade shall be not less than 0.014 inch. Each saw shall be furnished with a supplementary handle

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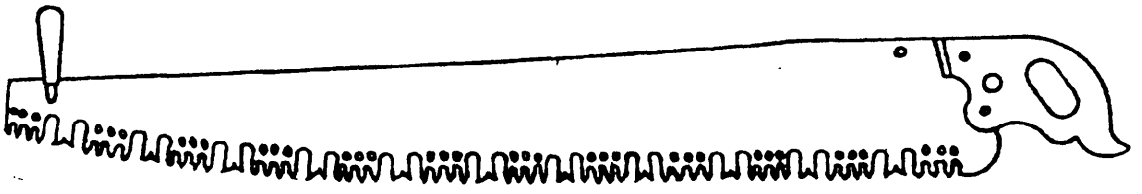


FIGURE 2.—Type I, crosscut saw, one-man.

which shall be easily and firmly attached without tools; it shall have at least 5 inches of gripping surface and be at least 1⅛ inches at the largest diameter. The saw shall be similar to figure 2 and shall conform to the dimensions shown in table I.

TABLE I.—*Type I, crosscut saws, one-man with style A or B teeth*

Size (over-all length of blade)	Thickness of blade		Width at butt		Width at point	
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Feet	*Inch*	*Inch*	*Inches*	*Inches*	*Inches*	*Inches*
3 ------	0.065	0.072	6	7¼	2⅛	2⅞
4½------	.065	.072	6¼	7¼	2¼	2⅞
4 ------	.065	.072	6¼	7¼	2¼	2⅞
4½------	.065	.072	6¼	7½	2½	2⅞

3.9 **Type II, two-man (see fig. 3).** — The saws shall be furnished with two handles which can be firmly secured to the blade without tools. Class 1 through class 4 saws shall be suitable for use with a loop-type or pin-type handle. Classes 5 and 6 shall be suitable for pin-type handles either with or without reversible guards.

3.9.1 *Handles.*—The handles shall be similar to the styles shown on figure 4 as specified (see 6.1). The loops or attaching bolts shall be of malleable iron or steel with high-tensile strength. The ferrules, clasps, washers, guards, and nuts shall be of cast iron or steel of substantial construction. The handles shall conform in length within a plus or minus ½ inch and in maximum thickness within plus or minus ⅛ inch with the dimensions shown on figure 4.

3.9.1.1 Loop-type handles shall be either style a or style b; the loop of style a handles shall be threaded to tighten on a nut embedded in the handle above the ferrule, the loop of style b handle shall extend the full length of the handles and engage in a metal cap at the opposite end. Style c pin-type handles shall be furnished with a metal face-

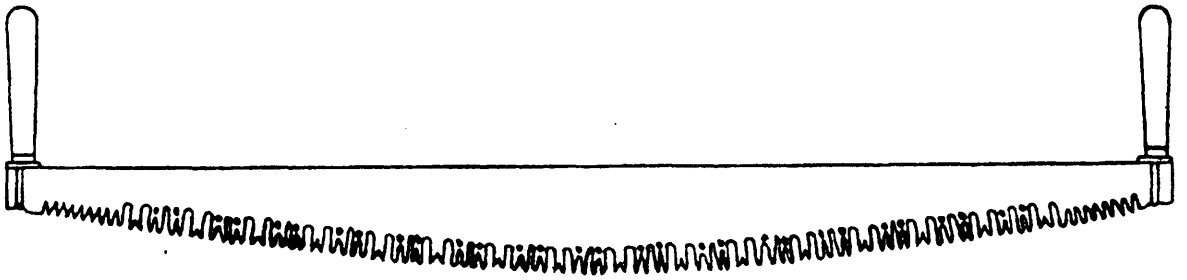
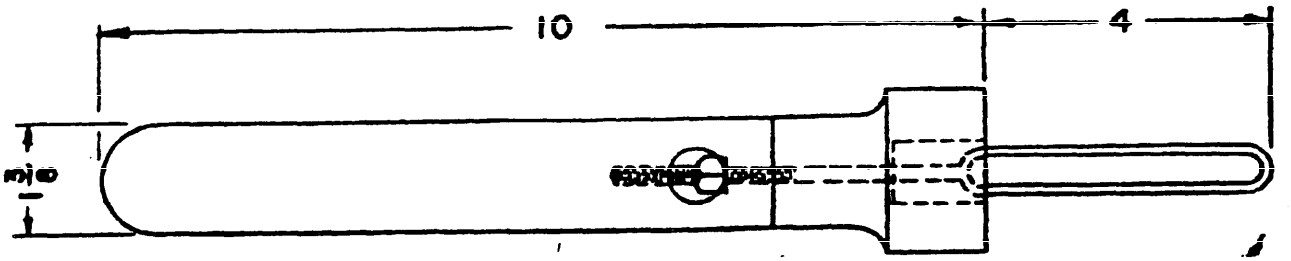
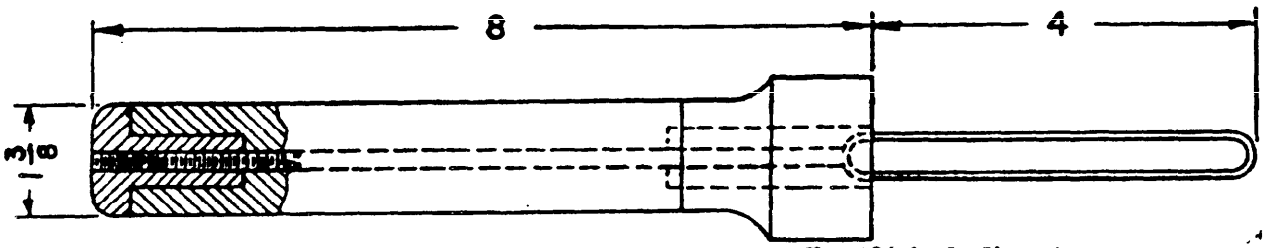


FIGURE 3.—Type II, crosscut saw, two-man.

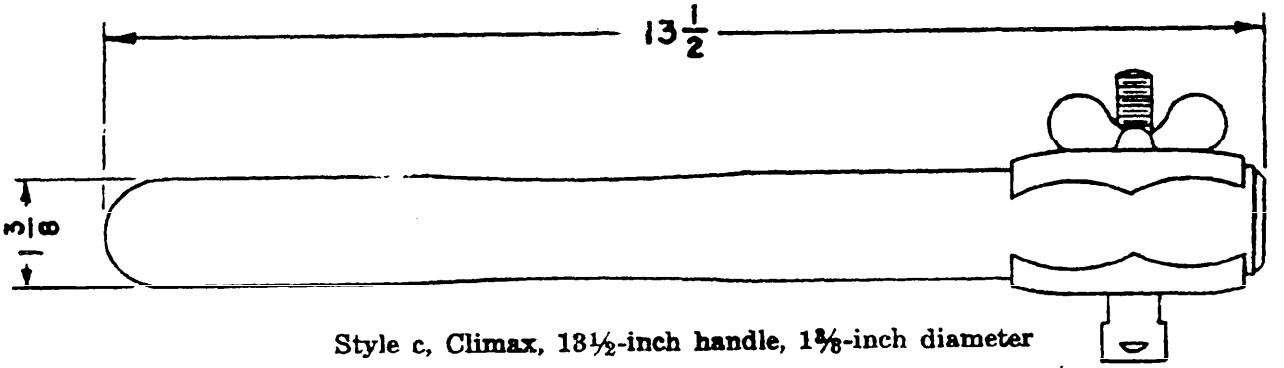
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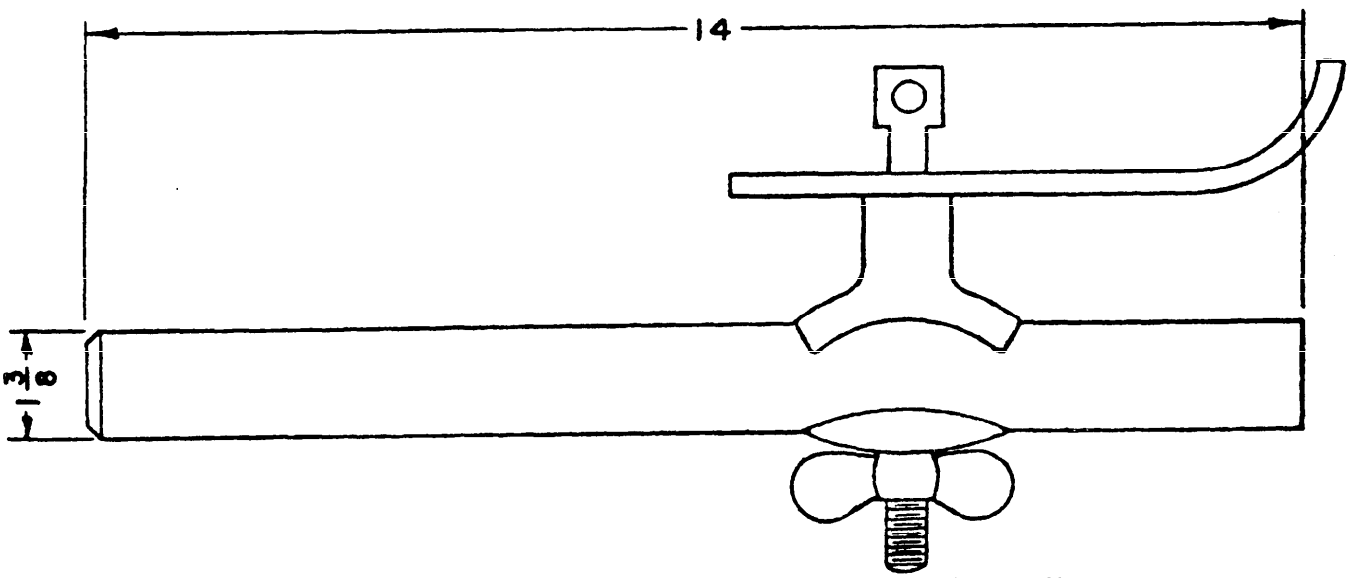
Style a, common loop, 4-inch loop, 10-inch handle, 1⅜-inch diameter



Style b, improved loop, 4-inch loop, 8-inch handle, 1⅜-inch diameter



Style c, Climax, 18½-inch handle, 1⅜-inch diameter



Style d, Reversible guard, Pacific type, 14-inch handle, 1⅜-inch diameter

FIGURE 4.—Patterns of handles for type II crosscut saws.

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plate, washer, malleable-iron or steel bolt and wing nut, commonly known as climax type. Style d, reversible guard, shall be of the pin type but so designed that it can be used in a vertical or horizontal position.

3.9.2 *Class 1, wide straight back.*—The saws shall be of the straight-back pattern with a wide blade. Saws shall have style A or style B teeth and shall conform to the dimensions shown in table II.

3.9.3 *Class 2, medium-wide straight back.*—The saws shall be similar to class 1 except for the dimensions. They shall conform to the dimensions shown in table III.

TABLE II.—*Class 1, wide straight-back, two-man crosscut saws*

Length on back	Thickness of cutting edge	Clearance on back of blade				Width of blade				Breast	
		At center		At ends		At center		At ends			
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Feet	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
5	0.083	0.038	0.051	0.015	0.028	6	6½	3	3½	2½	3½
5½	.083	.041	.051	.015	.028	6¼	6⅞	3	3½	2⅞	3¾
6	.083	.041	.051	.015	.028	6¾	7¼	3	3½	3⅜	4⅛

TABLE III.—*Class 2, medium-wide, two-man crosscut saws*

Length on back	Thickness of cutting edge	Clearance on back of blade				Width of blade				Breast	
		At center		At ends		At center		At ends			
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Feet	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
5	0.083	0.038	0.051	0.015	0.028	5	5¾	2¾	3½	1½	3½
5½	.083	.038	.051	.015	.028	5¼	6⅛	2⅞	3½	1¾	3¾
6	.083	.038	.051	.015	.028	5⅝	6⅜	3	3½	2	4¼
6½	.083	.038	.051	.015	.028	5¾	6⅝	3	3½	2¼	4¾

TABLE IV.—*Class 3, narrow skewback, two-man crosscut saws*

Length on back	Thickness of cutting edge	Clearance on back of blade				Width of blade				Breast	
		At center		At ends		At center		At ends			
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Feet	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
5	0.083	0.022	0.037	0.015	0.028	4¼	4½	3	3½	1¾	2¾
5½	.083	.022	.037	.015	.028	4⅜	4⅞	3	3½	2⅛	2¾
6	.083	.022	.037	.015	.028	4⅝	5	3	3½	2½	3¼

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3.9.4 *Class 3, narrow skewback.* — The saws shall have a curved back, and shall conform to the dimensions shown in table IV. Style 1 saws shall have style A or style B teeth, as specified.

3.9.5 *Class 4, uniform width.*—The saws shall be of uniform width the full length of the blade, and shall conform to the dimensions shown in table V. Saws shall have style A or style B teeth, as specified.

3.9.6 *Class 5, bucking saws (Pacific coast).*—The saws shall have style C teeth and shall be toothed to the end of the blade and adapted for use with styles c and d handles. The saws shall conform to the dimensions shown in table VI.

3.9.7 *Class 6, felling saws (Pacific coast).*—The saws shall be similar to class 5 with style C teeth and conform to the dimensions shown in table VII.

3.10 Finish.—The surface of the saw shall be polished in accordance with the best commercial practice for first-quality saws. There shall be no thinning or "rubbing off" near the edges of cutting teeth or rakers.

3.11 Workmanship. — The workmanship and appearance shall be first class and equal in every respect to the best commercial practice.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—All crosscut saws of the same type, class, tooth style, and size

TABLE V.—*Class 4, uniform width, two-man crosscut saws*

Length on back	Thickness of cutting edge	Clearance on back of blade				Width of blade		Breast	
		At center		At ends					
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Feet	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inches*	*Inches*	*Inches*
5	0.083	0.015	0.028	0.015	0.028	3½	3¾	2	3¼
5½	.083	.015	.028	.015	.028	3½	3¾	2¼	3½
6	.083	.015	.028	.015	.028	3½	3¾	2½	3¾

TABLE VI.—*Class 5, bucking, two-man crosscut saws*

Length on back	Thickness of cutting edge	Clearance on back of blade				Width of blade				Breast	
		At center		At ends		At center		At ends			
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Feet	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
5	0.095	0.034	0.040	0.020	0.026	6	6½	3¾	4½	1¾	2½
6	.095	.034	.040	.020	.026	6½	7¼	3¾	4½	2¼	3
6½	.095	.034	.040	.020	.026	6½	7⅞	3¾	4½	2⅝	3½
7	.095	.034	.040	.020	.026	7	7⅝	3¾	4½	2½	3¾

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TABLE VII.—*Class 6, felling, two-man crosscut saws*

Length on back	Thickness of cutting edge	Clearance on back of blade				Width of blade				Breast	
		At center		At ends		At center		At ends			
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Feet	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
5½	0.095	0.034	0.040	0.020	0.026	4⅝	5⅛	2⅝	3⅛	2⅛	3⅝
6	.095	.034	.040	.020	.026	5⅛	6	2⅝	3⅛	2⅛	3⅝
6½	.095	.034	.040	.020	.026	5⅛	6	2⅝	3⅛	3	3⅝
7	.095	.034	.040	.020	.026	5⅝	6⅛	2⅝	3⅛	3¼	4
7½	.095	.034	.040	.020	.026	5⅝	6⅛	2⅝	3⅛	3¼	4¼
8	.095	.034	.040	.020	.026	5⅛	6⅛	2⅝	3⅛	3¼	4¼
10	.095	.034	.040	.020	.026	5⅝	6⅛	2⅝	3⅛	2⅝	4

presented at one time shall be considered a lot for the purpose of acceptance inspection and tests.

4.1.2 *Sampling for inspection.*—A random sample of crosscut saws shall be selected from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the following single sampling inspection requirements in accordance with table VIII and Military Standard MIL-STD-105.

TABLE VIII. — *Sampling for visual and dimensional inspection A.Q.L. (approx.) — 1.5 percent defective*

Number of crosscut saws in inspection lot	Sample size, number of crosscut saws to be inspected (reference, Military Standard MIL-STD-105)
40 and under	Inpection Level III
41 to 300	Inspection Level II
301 and over	Inspection Level I

4.1.3 *Sampling for tests.*

4.1.3.1 Sampling for tests of tooth hardness (see 3.8) and bending tests shall be in accordance with table IX.

TABLE IX.—*Sampling for tests*

Number of crosscut saws in inspection sample	Number of crosscut saws to be tested	Acceptance number (defectives)	Rejection number (defective)
Under 25	3	0	1
25 to 49	5	0	1
50 and over	10	0	1

4.2 Inspection at the place of manufacture.

4.2.1 *Visual and dimensional inspection.*—Each of the sample crosscut saws selected in accordance with table VIII shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Dimensional checks shall include a check of straightness and uniform thickness (see 3.5 and 3.6). Any saw in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective saws in any sample exceeds the acceptance number for that sample the lot shall be rejected.

4.3 Testing at the place of manufacture.

4.3.1 *Hardness tests.*—The blades of the saws shall be tested for compliance with 3.3.

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4.3.1.1 *Tooth hardness.*—The suitability of the saw material shall be tested by filing. If the cutting teeth or rakers are excessively hard as not to permit filing, the lot of saws does not comply with this specification.

4.3.2 *Bending tests.*—Type II saws shall be tested by bending the saw to a U-shape between parallel surfaces spaced 18 inches apart for blades 0.083 inch thick and 21 inches apart for blades 0.095 inch thick. After bending, the permanent set shall not exceed 0.15 inch at the center of a 4-foot central-gage length.

4.3.3 If any of the test saws fail to meet the requirements for hardness or bending tests, the lot represented by the test samples shall be rejected.

4.4 Rejected lots may be offered again for Government inspection provided the contractor removes all nonconforming saws. The Government inspector shall again select and examine samples from such rejected lots to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

5.1 **Preservation.**—Cleaning and preservation shall be in accordance with the manufacturer's standard practice.

5.2 **Packaging.** — Commercial packages will be acceptable.

5.3 **Packing.**—The saw frames, packaged as specified in 5.2, shall be packed in accordance with the latest edition of the Consolidated Freight Classification Rules, Motor Freight Classification Rules, or Postal Regulations, whichever may be applicable. When fiberboard is used for container construction, the Mullen test shall be no less than 275 pounds. The gross weight of wooden boxes shall not exceed 150 pounds.

5.4 **Marking.**

5.4.1 *Interior containers.*—Packages and interior containers shall be marked with the name of the material and the name of the contractor or with a trade-mark of such known character that the source of supply may be readily determined.

5.4.2 *Exterior containers.*—Shipping containers shall be marked with the name of the item, the type and class, and the quantity therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 **Intended use.** — This specification covers one-man and two-man crosscut saws generally used for felling trees, cutting logs, and trimming large branches from trees.

6.2 **Ordering data.**—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following (see 1.2, 3.1.3, 3.9, 3.9.1, and 3.9.1.1):

a. Title, number, and date of this specification.
b. Type, class, and tooth style of saw required (see 1.2).
c. If styles of saws other than specified in 1.2 are required.
d. Style of handle (see 3.9.1.1).
e. Size of saw required (see tables I to VII, inclusive).
f. Whether domestic shipment immediate use; domestic shipment and storage; or overseas shipment is required (see section 5 and 7.2.3).

6.3 EXCEPT FOR DIMENSIONAL REQUIREMENTS SPECIFIED THEREIN THE ILLUSTRATIONS HEREIN ARE DESCRIPTIVE AND NOT RESTRICTIVE and are not intended to preclude the purchase of saws otherwise conforming to this specification.

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6.4 This specification supersedes GGG-S-61a in part as follows:

Type designation under this specification	*Former type designation under Specification GGG-S-61a*
Type I	Type L
Type II, class 1	Type M, class 1
Type II, class 2	Type M, class 2
Type II, class 3	Type M, class 3
Type II, class 4	Type M, class 4
Type II, class 5	Type M, class 5
Type II, class 6	Type M, class 6

6.5 **Transportation description.**—The applicable transportation description, carload and truckload minimum weights, is as follows:

Saws, hand, not otherwise indexed by name.
Carload weight 30,000 pounds.
Truckload weight 30,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specification and standard, of the issue in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 **Army, Navy, and Air Force.**

7.2.1 *Applicable specifications and standard.*

Military Specification:

MIL-P-15424—Packaging of Hand Tools; General Specification for Domestic and Overseas Preservation.

Military Standard:

MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Identification of product.*—In addition to the marking specified in 3.2, saws shall be marked in accordance with Military Standard MIL-STD-130.

7.2.3 *Preparation for delivery for domestic or overseas shipment.*—The saws shall be cleaned, preserved, unit protected, packaged, packed, and marked in accordance with Military Specification MIL-P-15424. The nomenclature shall be the exact nomenclature for the part or item that has been approved under the contract or order.

MILITARY INTEREST:
Army—E O Q Sig T
Navy—Sh MC Y
Air Force.

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